16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{ll} \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = & -0.2376 + 0.5311 DTIME \\ \text{(se)} & \quad (0.7505) \quad (0.2064) \end{array}$$

a. Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.

b. Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [*Hint*: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]

c. Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?

d. Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

a. DTIM = 1

$z = -0.2376 + 0.5311 \times 1 = 0.2935$

cdf logit : $P(AUTO=1) = \dfrac{e^{0.2935}}{1+e^{0.2935}} \approx 0.5728 = 57.28\%$

b. $P(AUTO=1) = \Phi(\beta_1 + \beta_2 DTIME) = -0.0644 + 0.3\,DTIME$

$P(AUTO=1, DTIME=1) = \Phi(2.9356) \approx 0.5931 = 59.31\%$

2者机率接近，相差 < 2%。因 logit, Probit 標準常態分配在数據中央，極为相似，不論用某个 model，对个体行为預測結果不会有顯著差異

c. $DTIME = 3,\ z = 1.3557$

$$\frac{d\hat{p}}{dx} = \frac{e^{1.3557}}{(1+e^{1.3557})^2} \times 0.5311 \approx 0.0865$$

d.

logit : $\dfrac{d\hat{p}}{dx} = \lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2 \cdot (-5))\tilde{\gamma}_2 = 0.0262$

probit : $\dfrac{d\hat{p}}{dx} = \phi(\tilde{\gamma}_1 + \tilde{\gamma}_2 \cdot (-5))\tilde{\gamma}_2 = 0.0352$

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is $P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The maximum likelihood estimates of the parameters are $\tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021x$. The maximized value of the log-likelihood function is -1.612.

a. Calculate the probability that $y = 1$ if $x = 1.5$, given the values of the maximum likelihood estimates.

b. Using the threshold 0.5 and the result in part (a), predict the value of y if $x = 1.5$, the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome $y = 1$ in the first observation.

c. Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of $\Phi(\bullet)$ and using the given $N = 3$ data pairs, if the parameter values are $\gamma_1 = -1$ and $\gamma_2 = 2$. Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?

d. For the logit model, the value of the likelihood function (16.14), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be between zero and one. True or false? Explain.

e. For the logit model, the value of the log-likelihood function (16.15), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be negative. True or false? Explain.

a. $z = -1.836 + 1.5 \times 3.021 = 2.6955$

$\Lambda(2.9655) = \dfrac{1}{1+e^{-2.9655}} \approx 0.9368$

b. $\because 0.9368 > 0.05$ $\therefore$ 預測的 $\hat{y}=1$ 会是正確的

c. $r_1 = -1$, $r_2 = 2$

算出來的 < -1.612

d.e. True

概似函數 L 是多个机率值連乘 $(0 \le p \le 1) \rightarrow$ 其值必落入 $[0,1]$

$\ln(L) \le 0$ 又 $0 \le L \le 1$，除了 $\ln(1) = 0$，其他皆為負

16.18 Mortgage lenders are interested in determining borrower and loan characteristics that may lead to delinquency or foreclosure. In the data file *lasvegas* are 1000 observations on mortgages for single family homes in Las Vegas, Nevada during 2008. The variable of interest is *DELINQUENT*, an indicator variable = 1 if the borrower missed at least three payments (90 + days late), but 0 otherwise. Explanatory variables are *LVR* = the ratio of the loan amount to the value of the property; *REF* = 1 if purpose of the loan was a "refinance" and = 0 if loan was for a purchase; *INSUR* = 1 if mortgage carries mortgage insurance, 0 otherwise; *RATE* = initial interest rate of the mortgage; *AMOUNT* = dollar value of mortgage (in \$100,000); *CREDIT* = credit score, *TERM* = number of years between disbursement of the loan and the date it is expected to be fully repaid, *ARM* = 1 if mortgage has an adjustable rate, and = 0 if mortgage has a fixed rate.

a. Estimate the linear probability (regression) model explaining *DELINQUENT* as a function of the remaining variables. Use White heteroskedasticity robust standard errors. Are the signs of the estimated coefficients reasonable?

b. Use logit to estimate the model in (a). Are the signs and significance of the estimated coefficients the same as for the linear probability model?

c. Compute the predicted value of *DELINQENT* for the 500th and 1000th observations using both the linear probability model and the logit model. Interpret the values.

d. Construct a histogram of *CREDIT*. Using both linear probability and logit models, calculate the probability of delinquency for *CREDIT* = 500, 600, and 700 for a loan of \$250,000 (*AMOUNT* = 2.5). For the other variables, let the loan to value ratio (*LVR*) be 80%, the initial interest rate is 8%, all indicator variables take the value 0, and *TERM* = 30. Discuss similarities and differences among the predicted probabilities from the two models.

e. Using both linear probability and logit models, compute the marginal effect of *CREDIT* on the probability of delinquency for *CREDIT* = 500, 600, and 700, given that the other explanatory variables take the values in (d). Discuss the interpretation of the marginal effect.

f. Construct a histogram of *LVR*. Using both linear probability and logit models, calculate the probability of delinquency for *LVR* = 20 and *LVR* = 80, with *CREDIT* = 600 and other variables set as they are in (d). Compare and contrast the results.

g. Compare the percentage of correct predictions from the linear probability model and the logit model using a predicted probability of 0.5 as the threshold.

h. As a loan officer, you wish to provide loans to customers who repay on schedule and are not delinquent. Suppose you have available to you the first 500 observations in the data on which to base your loan decision on the second 500 applications (501–1,000). Is using the logit model with a threshold of 0.5 for the predicted probability the best decision rule for deciding on loan applications? If not, what is a better rule?